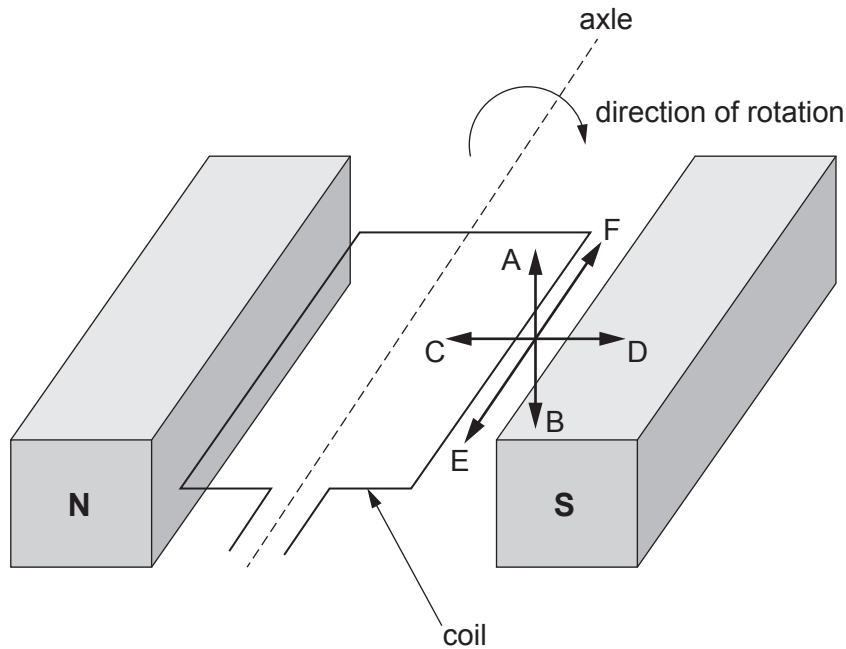


5. The operation of electric generators can be explained by electromagnetic induction.

(a) The diagram shows part of an a.c. generator.



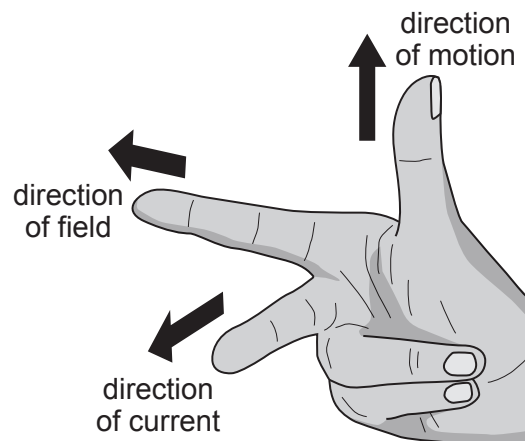
(i) State the direction (A, B, C, D, E or F) in which the magnetic field acts. [1]

Direction = .....

(ii) State the direction (A, B, C, D, E or F) in which the right hand side of the coil moves. [1]

Direction = .....

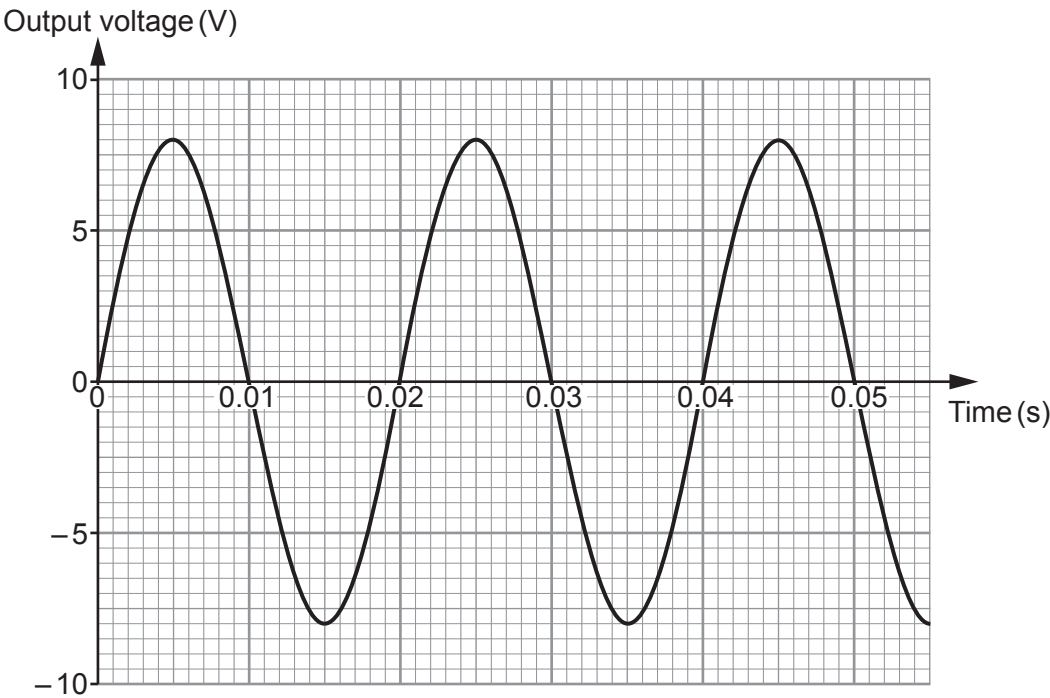
(iii) Use Fleming's right hand rule to determine the direction (A, B, C, D, E or F) of the induced current in the right hand side of the coil. [1]



Direction = .....



(iv) When the coil spins, a voltage is produced.  
This is shown on the graph.



I. Use the graph to find the maximum voltage produced by the generator. [1]

maximum voltage = ..... V

II. Use the graph to find the time for one rotation (spin) of the coil. [1]

time = ..... s

III. **Complete the table** below to show the effect of each change made.  
Use the words **increases**, **decreases** or **no effect**.  
Three boxes have been completed for you. [3]

| Change made            | Effect on maximum voltage | Effect on time for 1 rotation (spin) |
|------------------------|---------------------------|--------------------------------------|
| Stronger magnets       | increases                 | no effect                            |
| More turns on the coil | .....                     | no effect                            |
| Coil spins slower      | .....                     | .....                                |



(b) Transformers also work by electromagnetic induction.

(i) Tick (✓) the boxes next to the **three** correct statements about transformers. [3]

Transformers change a.c. to d.c.

☐

Transformers change d.c. voltages.

☐

The core is made from iron.

☐

There is an electric current in the core.

☐

The primary coil creates an alternating magnetic field.

☐

Transformers can increase or decrease voltages.

☐

(ii) A transformer increases voltage from 12 V to 36 V.  
The primary coil contains 20 turns of wire.  
Use the equation:

$$N_2 = N_1 \times \frac{V_2}{V_1}$$

to calculate the number of turns in the secondary coil,  $N_2$ . [2]

$N_2 = \dots\dots\dots$

